University of Florida, Department of Mathematics

Variational Analysis/Numerical Optimization PhD Exam, August 2025

Do 8 problems, 4 from problems 1–5 and 4 from problems 6–10.

1. Recall that $f : \mathbb{R}^n \to \mathbb{R}$ is convex with modulus $\theta \geq 0$ if and only if

$$f[(1-\alpha)\mathbf{x} + \alpha\mathbf{y}] \leq (1-\alpha)f(\mathbf{x}) + \alpha f(\mathbf{y}) + \frac{\theta\alpha(\alpha-1)}{2}\|\mathbf{y}-\mathbf{x}\|^2$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha \in [0,1]$.

(a) If f is continuously differentiable, then show that f is convex with modulus $\theta \geq 0$ if and only if

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})(\mathbf{y}-\mathbf{x}) + \frac{\theta}{2}\|\mathbf{y}-\mathbf{x}\|^2$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

(b) If f is twice continuously differentiable, then show that f is convex with modulus $\theta \geq 0$ if and only if the smallest eigenvalue of $\nabla^2 f$ is $\geq \theta$.

2. Suppose $f : \mathcal{K} \to \mathbb{R}$ where $\mathcal{K} \subset \mathbb{R}^n$ is a convex set.

(a) If $\mathbf{x}^*$ is a local minimizer of f, then show that

$$\nabla f(\mathbf{x}^*)(\mathbf{x}-\mathbf{x}^*) \geq 0 \qquad \text{for all } \mathbf{x} \in \mathcal{K}. \tag{1}$$

(b) If f is convex and differentiable at some point $\mathbf{x}^* \in \mathcal{K}$ where (1) holds, then show that $\mathbf{x}^*$ is a global minimizer for f over $\mathcal{K}$. Moreover, if the modulus of convexity of f is strictly greater than 0, then $\mathbf{x}^*$ is the unique global minimizer of f over $\mathcal{K}$.

(c) Consider the quadratic programming problem

$$\min \frac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} - \mathbf{b}^{\mathsf{T}}\mathbf{x} \qquad \text{subject to } \mathbf{x} \in \mathcal{K},$$

where $\mathcal{K} \subset \mathbb{R}^n$ is a closed convex set and $\mathbf{A} \in \mathbb{R}^{n\times n}$ is positive definite with smallest eigenvalue $\alpha > 0$. Show that the solutions $\mathbf{u}_i$, $i = 1, 2$, associated with $\mathbf{b} = \mathbf{b}_i$, $i = 1, 2$, respectively, satisfy

$$\|\mathbf{u}_1 - \mathbf{u}_2\| \leq \|\mathbf{b}_1 - \mathbf{b}_2\|/\alpha.$$

3. Let $\mathbf{U}$ and $\mathbf{V} \in \mathbb{R}^{n\times m}$ and $\mathbf{M} \in \mathbb{R}^{n\times n}$ with $\mathbf{M}$ invertible.

(a) If $\mathbf{I} + \mathbf{V}^{\mathsf{T}}\mathbf{M}^{-1}\mathbf{U}$ is invertible, then show that $\mathbf{M} + \mathbf{U}\mathbf{V}^{\mathsf{T}}$ is invertible with

$$(\mathbf{M} + \mathbf{U}\mathbf{V}^{\mathsf{T}})^{-1} = \mathbf{M}^{-1} - \mathbf{M}^{-1}\mathbf{U}(\mathbf{I} + \mathbf{V}^{\mathsf{T}}\mathbf{M}^{-1}\mathbf{U})^{-1}\mathbf{V}^{\mathsf{T}}\mathbf{M}^{-1}.$$

(b) Consider the quadratic program

$$\min \frac{1}{2}\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} - \mathbf{b}^{\mathsf{T}}\mathbf{x} \qquad \text{subject to } \mathbf{a}^{\mathsf{T}}\mathbf{x} = c, \tag{2}$$

where $\mathbf{A}$ is a symmetric, positive definite matrix. The penalty approximation to (2) is

$$\min \frac{1}{2}\mathbf{x}^\mathsf{T}\mathbf{A}\mathbf{x} - \mathbf{b}^\mathsf{T}\mathbf{x} + \frac{p}{2}(\mathbf{a}^\mathsf{T}\mathbf{x} - c)^2,$$

where $p > 0$ is the penalty parameter. Use the formula from (a) to obtain an explicit solution to the penalized problem in terms of $\mathbf{A}^{-1}$, and the parameters of the quadratic program.

4. Consider the primal optimization problem

$$\inf_{\mathbf{x}\in\mathcal{K}} \{f(\mathbf{x}) : \mathbf{h}(\mathbf{x}) = 0, \quad \mathbf{g}(\mathbf{x}) \le 0\}, \tag{P}$$

where $\mathcal{K} \subset \mathbb{R}^n$, $\mathbf{h} : \mathbb{R}^n \to \mathbb{R}^m$, and $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^l$. The dual of (P) is

$$\sup_{\substack{\boldsymbol{\mu}\ge 0\\ \boldsymbol{\lambda}\in\mathbb{R}^m}} L(\boldsymbol{\lambda}, \boldsymbol{\mu}), \qquad \text{where} \qquad L(\boldsymbol{\lambda}, \boldsymbol{\mu}) := \inf_{\mathbf{x}\in\mathcal{K}} f(\mathbf{x}) + \boldsymbol{\lambda}^\mathsf{T}\mathbf{h}(\mathbf{x}) + \boldsymbol{\mu}^\mathsf{T}\mathbf{g}(\mathbf{x}). \tag{D}$$

(a) Show that the dual objective L is a concave function.

(b) Suppose that $\mathcal{K} = \mathbb{R}^n$, $\mathbf{h}(\mathbf{x}) = \mathbf{A}\mathbf{x} - \mathbf{b}$ where $\mathbf{A} \in \mathbb{R}^{m\times n}$, each component of $\mathbf{g}$ is convex, and there exists a solution $\mathbf{x}^*$ to (P) where the first-order optimality conditions are satisfied. Show that there is no duality gap. That is, show that the infimum in (P) is equal to the supremum in (D).

5. Consider the following primal quadratic program:

$$\min q(\mathbf{x}) := \frac{1}{2}\mathbf{x}^\mathsf{T}\mathbf{x} - \mathbf{b}^\mathsf{T}\mathbf{x} \qquad \text{subject to } \mathbf{a}^\mathsf{T}\mathbf{x} = c, \ \mathbf{x} \ge 0, \tag{QP}$$

where $\mathbf{b}$ and $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a} > 0$, and $c > 0$ is a scalar. An associated dual program is

$$\max_{\lambda} L(\lambda) := \min_{\mathbf{x}\ge 0} q(\mathbf{x}) + \lambda(\mathbf{a}^\mathsf{T}\mathbf{x} - c). \tag{QD}$$

Since λ is a scalar, the dual problem (QD) is much easier to solve than the primal problem (QP).

(a) The minimization over $\mathbf{x} \ge 0$ in the dual function L uncouples into n independent minimizations over each component x_i. Give a formula for $x_i(\lambda)$, the minimizer over $x_i \ge 0$ of the dual function at λ.

(b) The points where $\lambda = b_i/a_i$, $1 \le i \le n$, are known as the break points of the dual function. Give a formula for the value of the dual function and its derivative at any point λ which is not a break point. Show that $L'(\lambda) = \mathbf{a}^\mathsf{T}\mathbf{x}(\lambda) - c$.

(c) Show that L' is a decreasing function of λ (recall that L is concave), and $L'(\lambda)$ is linear in λ between the break points. Based on these observations, give an algorithm for maximizing L.

(d) Given a solution λ^* of the dual problem, show that $\mathbf{x}(\lambda^*)$ is the solution of the primal problem.

6. Let us consider the minimization problem

$$\min I(y) := \int_0^1 f(y'(x), y(x), x)\, dx \qquad \text{subject to } y \in C_0^1, \tag{3}$$

where f is a given continuously differentiable function. If y^* is a local minimizer of this problem, s is a scalar, $h \in C_0^1$, then setting to zero the derivative of $I(y^* + sh)$ evaluated at $s = 0$ yields the following first-order optimality condition:

$$\int_0^1 \left[\left(\frac{\partial f}{\partial y'}\right)^*(x)h'(x) + \left(\frac{\partial f}{\partial y}\right)^*(x)h(x)\right] dx = 0 \tag{4}$$

for all $h \in C_0^1$, where

$$\left(\frac{\partial f}{\partial y'}\right)^*(x) := \left(\frac{\partial f}{\partial y'}\right)^*(y^{*'}(x), y^*(x), x) \quad \text{and} \quad \left(\frac{\partial f}{\partial y}\right)^*(x) := \left(\frac{\partial f}{\partial y}\right)^*(y^{*'}(x), y^*(x), x).$$

(a) As a consequence of (4), show that

$$\left(\frac{\partial f}{\partial y'}\right)^* \in C^1 \tag{5}$$

and the Euler equation is satisfied. Note that it is only assumed that f is continuously differentiable; you should show that at a local minimizer of (3), the partial derivative in (5) is continuously differentiable.

(b) For the special case

$$f(y'(x), y(x), x) = \frac{1}{2}y'(x)^2 - \phi(x)y(x)^2,$$

where $\phi \in C^k$ for some integer $k \geq 0$, what does the observation in (a) imply concerning the smoothness of a minimizer y^* for (3).

7. Consider the equation

$$(P(x)u'(x))' = Q(x)u(x), \qquad x \in [0,1], \tag{E}$$

where Q is continuous and P is continuously differentiable and strictly positive on $[0,1]$. Suppose (E) has the property that whenever u is a solution of (E) with $u(0) = u(a) = 0$ for some $0 < a \leq 1$, u is identically zero on $[0,a]$. Show that the solution of the initial-value problem for (E), with the initial conditions $u(0) = 0$ and $u'(0) = 1$, is strictly positive on $(0,1]$.

8. Consider the variational problem

$$\min \int_0^1 \left[\frac{1}{2}u'(x)^2 + e^{u(x)}\right] dx \qquad \text{subject to } u \in \mathcal{H}_0^1.$$

(a) What is the first-order necessary optimality condition (Euler equation) for this variational problem?

(b) Consider a uniform mesh $x_k = kh$ where $h = 1/N$; let u_k denote the approximation to $u(x_k)$. Of course, $u_0 = u_N = 0$. Give a finite difference approximation in terms of $u_1, \dots, u_{N-1}$ to the solution of the Euler equation.

(c) Let $\mathbf{F}(\mathbf{u}) = 0$ denote the finite difference system where $\mathbf{u} = (u_1, u_2, \dots, u_{N-1})^\mathsf{T} \in \mathbb{R}^{N-1}$, and let $\mathbf{u}^*$ denote the vector formed by evaluating the solution of the Euler equation at the mesh points $x_1, x_2, \dots, x_{N-1}$. Obtain a bound for the components of $\mathbf{F}(\mathbf{u}^*)$ in terms of the mesh spacing h.

9. Consider the initial-value problem

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{u}(t), \qquad \mathbf{x}(0) = 0,$$

where $\mathbf{x} : [0,1] \to \mathbb{R}^n$ and $\mathbf{A} \in L^\infty([0,1]; \mathbb{R}^{n\times n})$.

(a) Assuming $\mathbf{u} \in L^2$, obtain a bound for the sup-norm of $\mathbf{x}$ in terms of the L^2 norm of $\mathbf{u}$.

(b) Let $\Omega : \mathcal{H}_0^1 \to \mathbb{R}$ be defined by

$$\Omega(h) = \frac{1}{2}\int_0^1 r^2(x)\, dx \qquad \text{where } r(x) = h'(x) + w(x)h(x),$$

where w is uniformly bounded on $[0, 1]$. Show that there exists $\beta > 0$ such that

$$\Omega(h) \geq \beta(\|h\|^2 + \|h'\|^2) \qquad \text{for all } h \in \mathcal{H}_0^1,$$

where $\|\cdot\|$ is the L^2 norm.

10. Consider the following control problem:

$$\min \int_0^1 \left[\frac{1}{2}u^2(x) + y(x)\right] dx \qquad \text{subject to } y'(x) = u(x),\ y(0) = 0,\ u(x) \geq \ell(x),$$

where $\ell(x)$ is a given lower bound for the control $u(x)$ at each $x \in [0, 1]$.

(a) What is the first-order optimality condition (Pontryagin minimum principle) for this problem?

(b) What is the solution of the control problem?